Installation Procedure (v1.2.1)
Workshop: **STM32C5 Workshop**

Please follow all the installation steps below before the Workshop.

Workshop – Requirements

Important:

You need to have administration rights to be able to install drivers and software and to do the workshop.

System requirements:

Windows® (10 or later, 64 bits (x64)), macOS® (15 or 26), or Linux® (Ubuntu® 22.04, Ubuntu® 24.04, Ubuntu® 26.04, Fedora® 43 and Fedora® 44.). Please see the STM32CubeMX2 installation guide for more details, it can be found at:
(https://dev.st.com/stm32cube-docs/stm32cubemx2/1.1.2/en/docs/markup/CubeMX2_How_To_Start/CubeMX2_Installation.html)

Minimum Hardware Requirements:

- 1 USB port
- Nucleo-C562RE
- 1 USB Type-C Cables
- 1 M2M jumper cable to change ADC value
- 1 F2F jumper cable for FDCAN connection
- 4 GB of system memory (RAM)
- 7 GB of available disk space

Important Notes:

1. Make sure the system's USB ports to be used in the workshop are enabled.

STM32C5 Workshop - Installation Procedure – Introduction:

The following Software/Tools/Libraries are needed for the Workshop:

1. STM32CubeMX2: version 1.1.2 is required, if you have a previous version you will need to install a new version as explained in this installation procedure.

2. VSCode with STM32CubeIDE Extension

3. STM32CubeProgrammer: version 2.23.0 minimum required, if you have a previous version you will need to install a new version as explained in this installation procedure.

4. Presentation and code to use during the workshop: It can be found in a downloadable zip file mentioned on the page below.

Notes:

1- **Some of the versions of the software and libraries on the ST website might have newer versions than the version seen in the document, please use the latest version from the links provided in the installation procedure below.**
2- **The instructions and screenshots are specific to tools versions specified above, the layout and appearance might change with version but the procedural steps for the workshop would remain the same.**
3- **The instructions and screenshots are specific to a Windows® based system.**
4- **The look of the website may change but the links provided in this document remain the same.**

The installation procedure can take several hours for downloading and installing based on the speed of the internet connection and performance of the computer system.

Questions and support ahead of the workshop

1. If you have issues during the software download and install, contact ST by entering an Online Support Request for workshops at https://ols.st.com/s/newcase?o=ws to help resolve the issue.

2. In the Workshop/Event Description field, select:
   "**STM32C5 MCU workshop**"

3. To ensure your request is quickly routed to the correct support team, please indicate the Workshop Request Type, either Technical or Non-Technical, that best describes your question.

STM32C5 Workshop - Installation Procedure

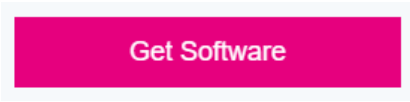
For the Workshop installation please follow the steps below:

1- STM32CubeMX2: STM32Cube initialization code generator

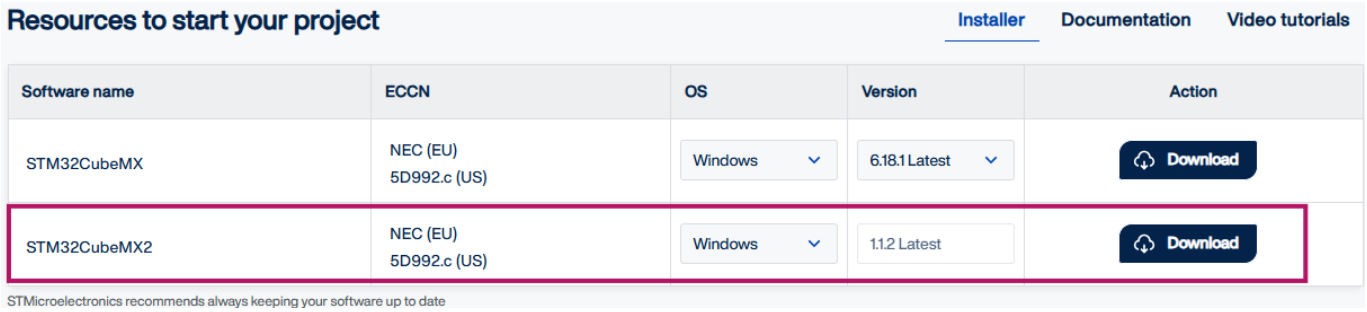
Click Link: STM32CubeMX | Software - STMicroelectronics
Minimum version to be installed: 1.1.2
Or use Direct Install Link:
https://www.st.com/en/development-tools/stm32cubemx.html

In the web browser, a similar page as follows will come up:

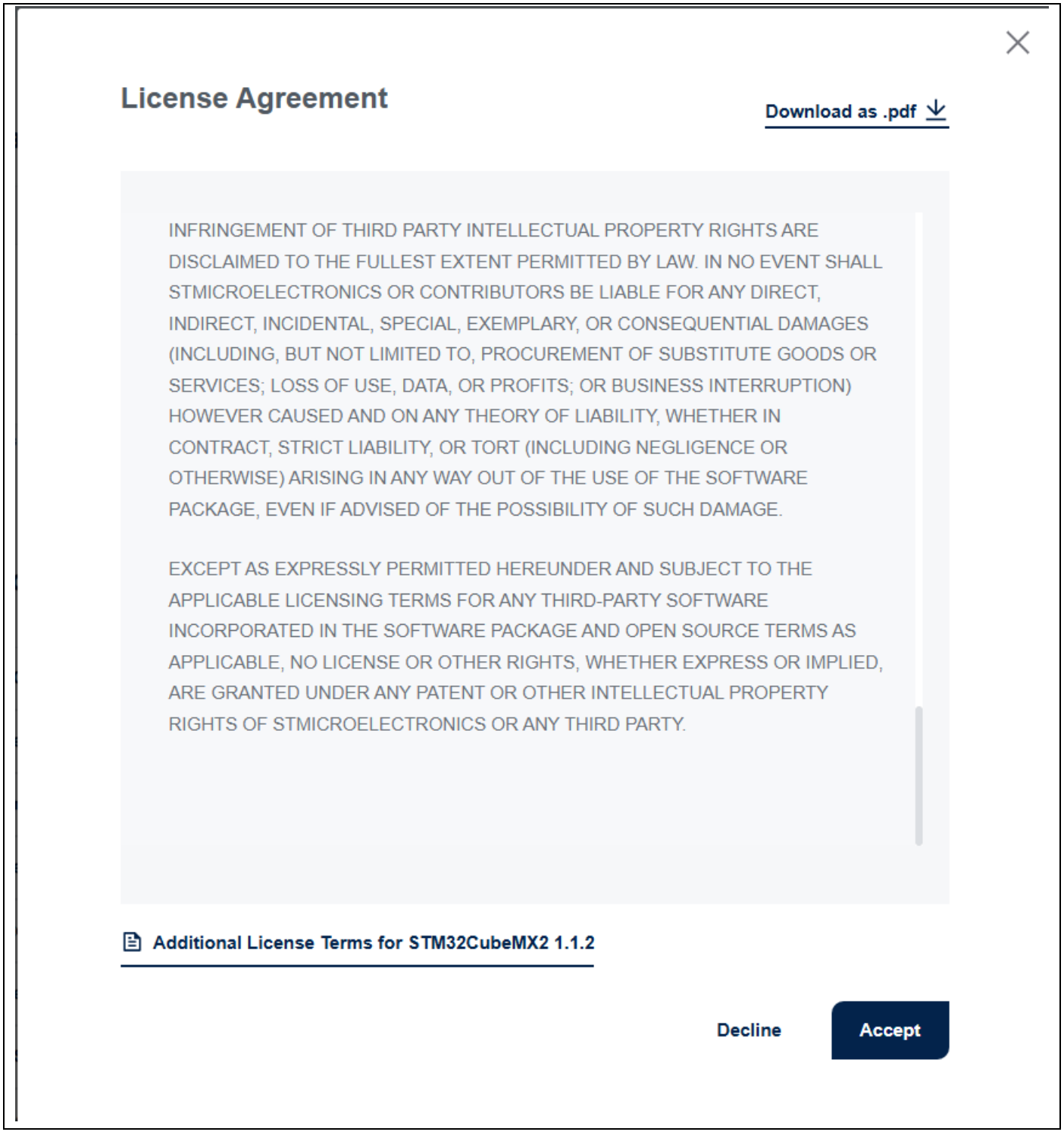


Click on “Get Software” button: 

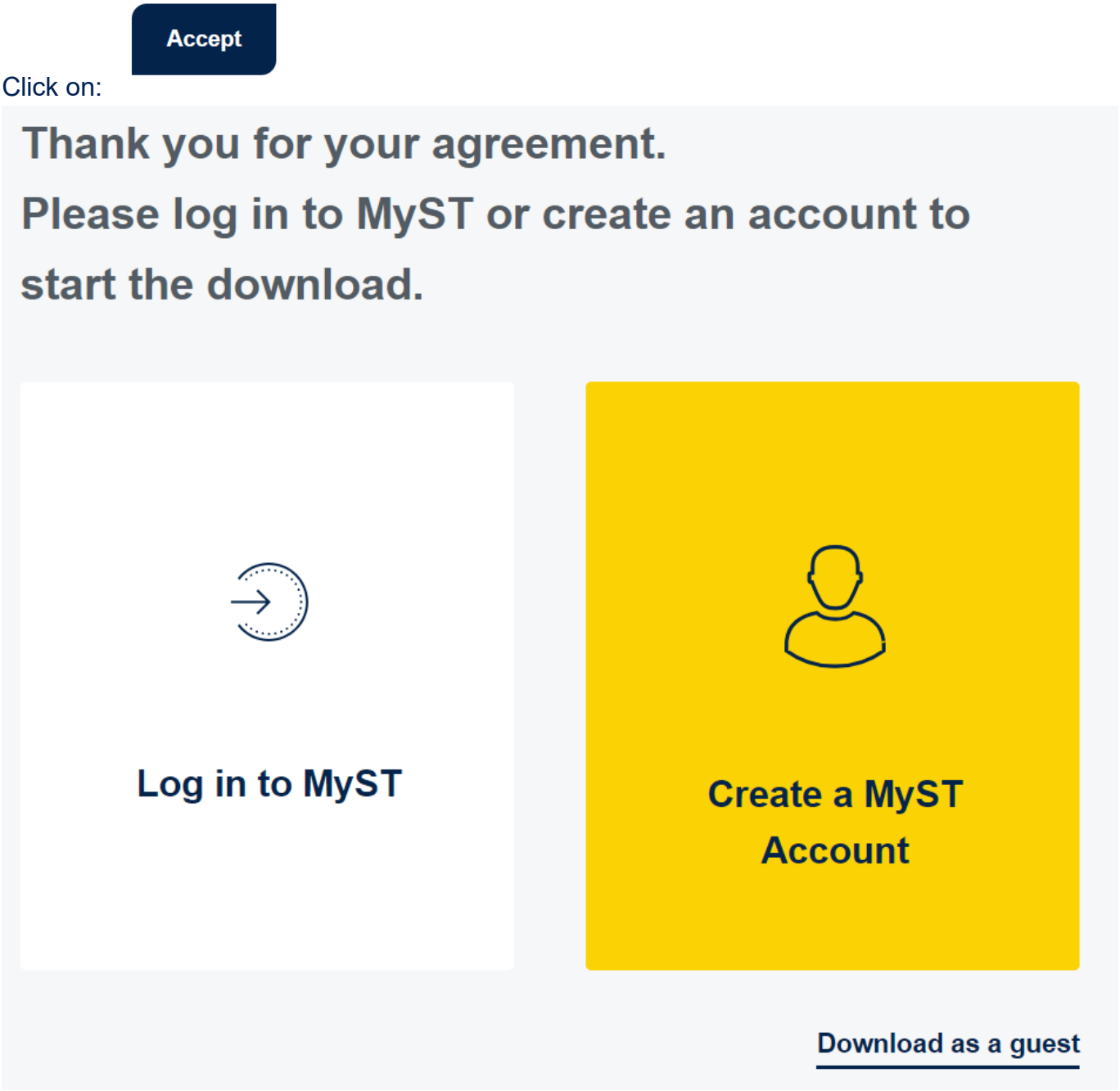
Click on “Download” for the latest version for your target operating system (shown below is for Windows)



When the following page appears:



Click on:



There are 3 ways of getting the software:

Log in to MyST

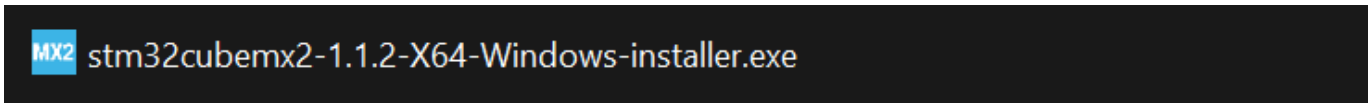
If you have an ST account, click on:

If you don’t have one, create one by clicking on: **Create a MyST Account**

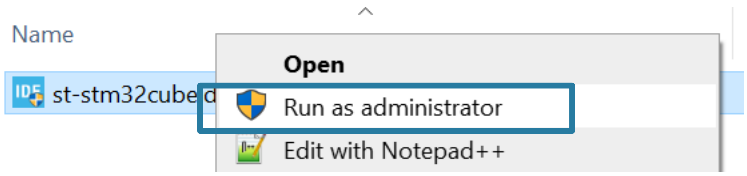
If you don’t have an account and don’t want to create one, use “Download as a guest option”

The software will automatically download in your browser.

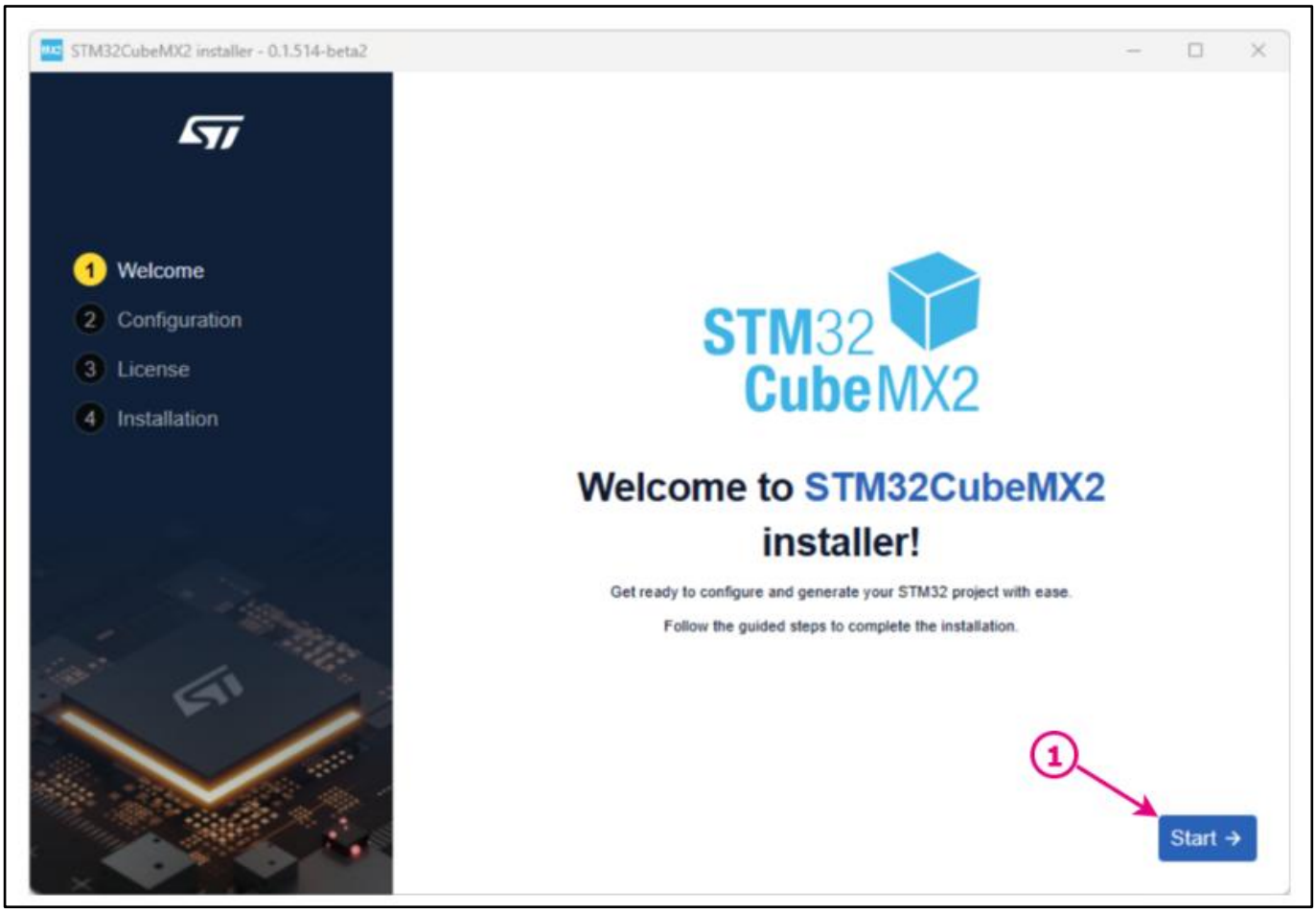
If not click “Get Software” Get Software



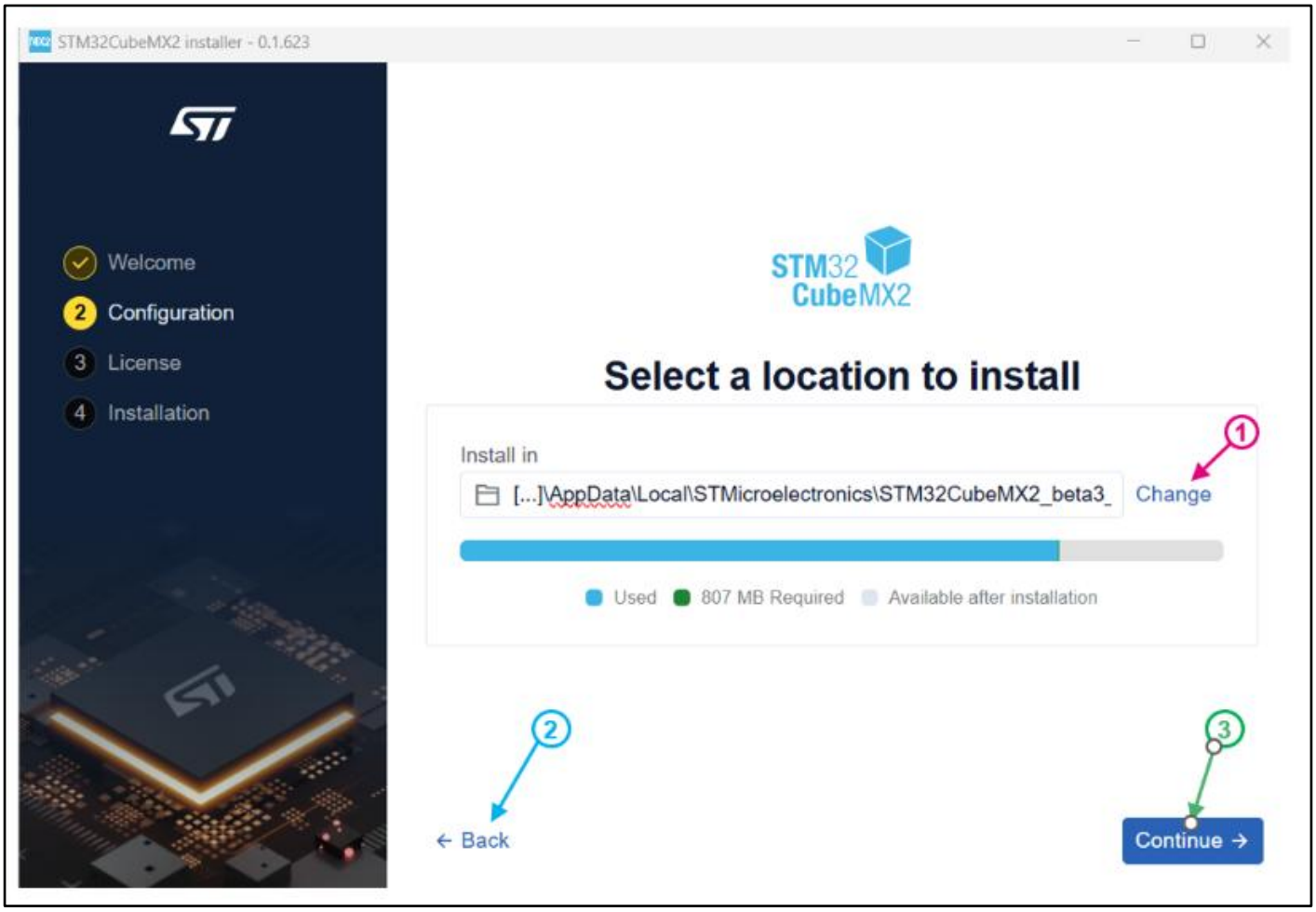
Right Click on ‘stm32cubemx2-1.1.2-X64-Windows-installer.exe’ and click on ‘Run as administrator’:



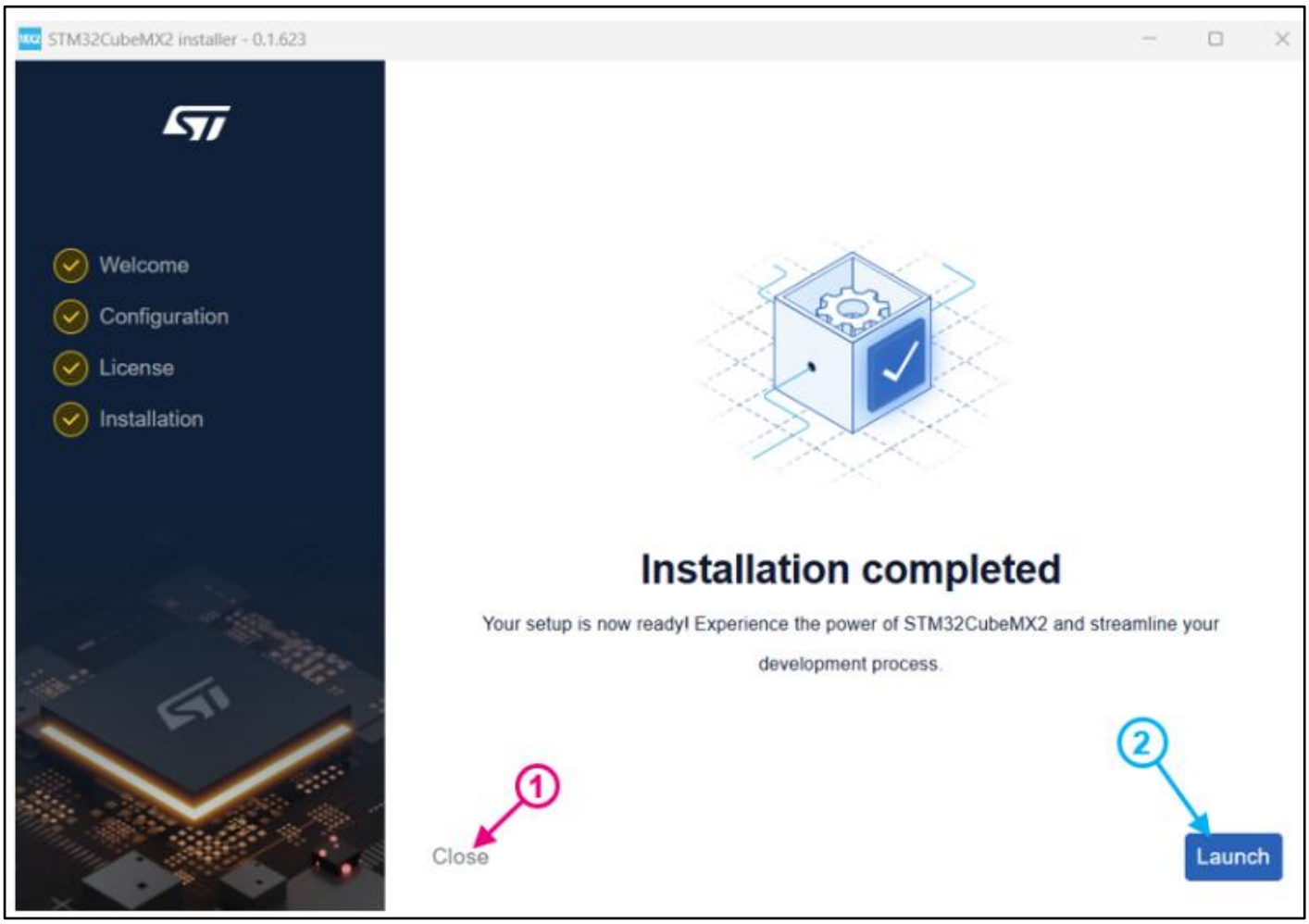
Press “Start”:



Press “Continue”:



Once installation is complete, “Launch” the tool

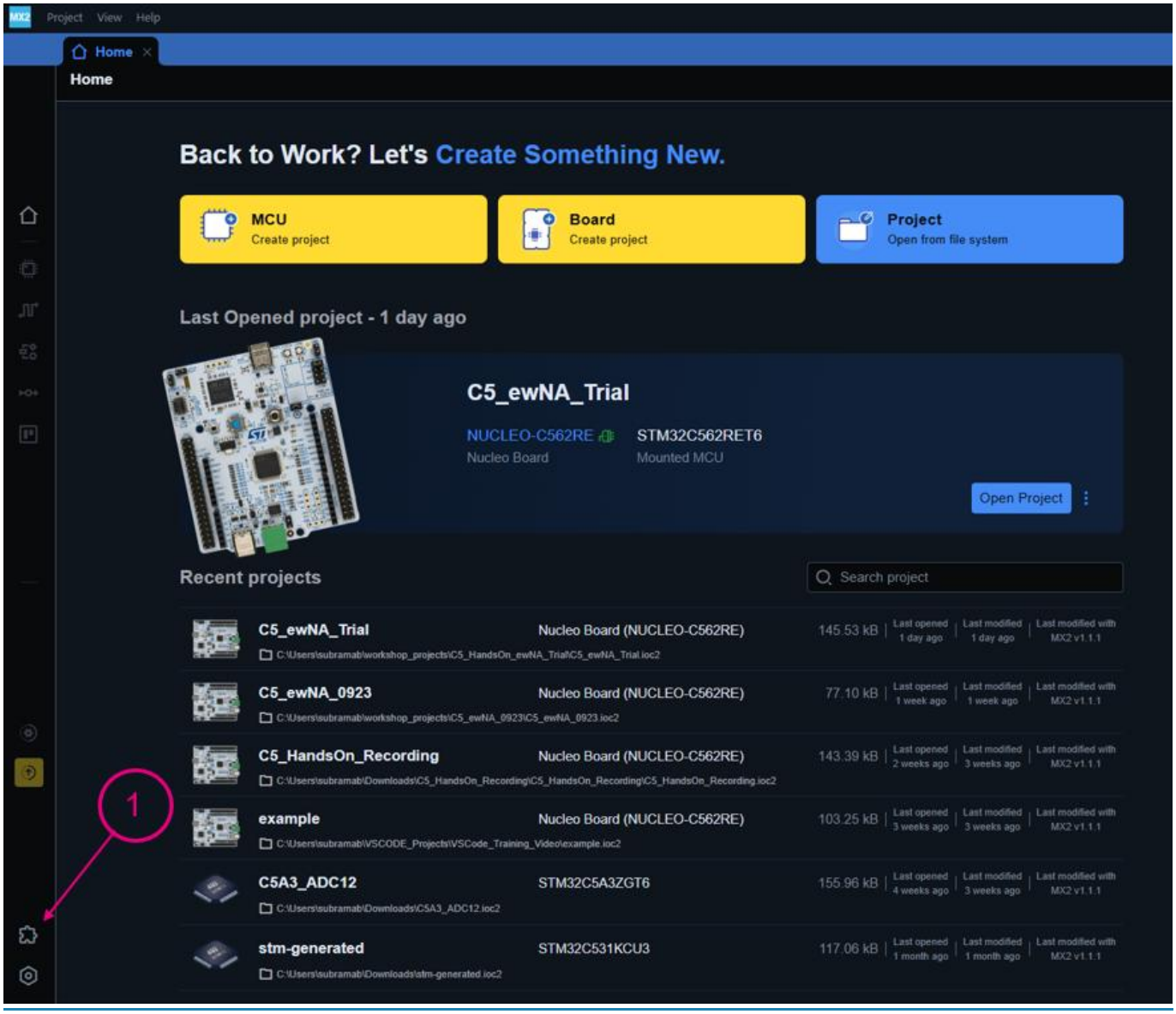


Downloading STM32CubeMX2 packs

Before creating a STM32CubeMX2 project, you will need to install the necessary STM32CubeMX2 packs. This can be done in three different ways:

1. From Pack Manager

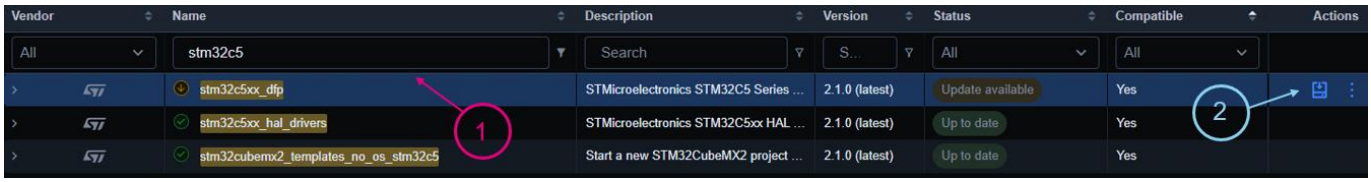
From within the application, navigate to the Pack Manager in the bottom right of the page:



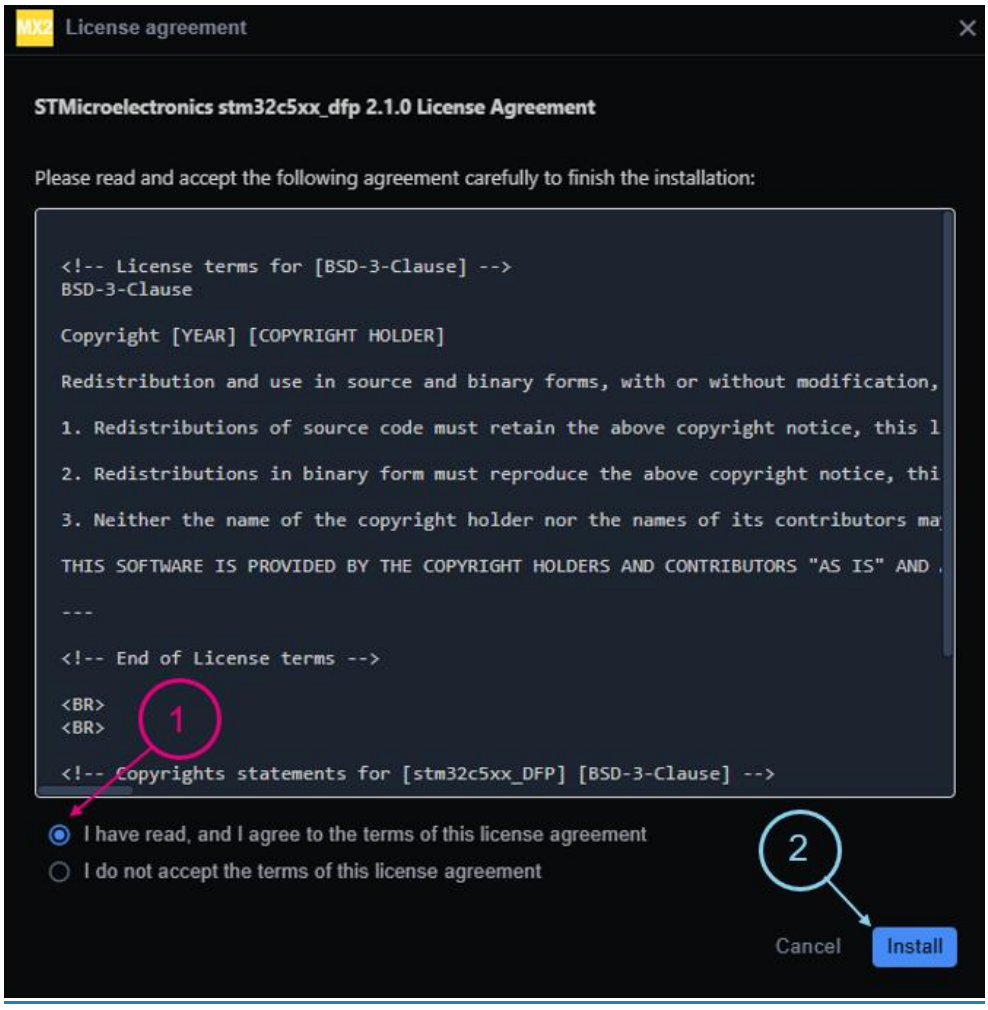
From within the Pack manager, download the following packs

- stm32c5xx_dfp
- stm32c5xx_hal_drivers
- stm32cubemx2_templates_no_os_stm32c5
- Syscalls V2.0.0
- CMSIS

Search for the packs using the top search bar, and click on the install button to the right.



Accept the terms of the license agreement and click **Install** on the bottom right.



2. From Command Line

You can use the following commands to install the necessary packages through the CLI.

>cube mx start
>timeout /t 5 >nul

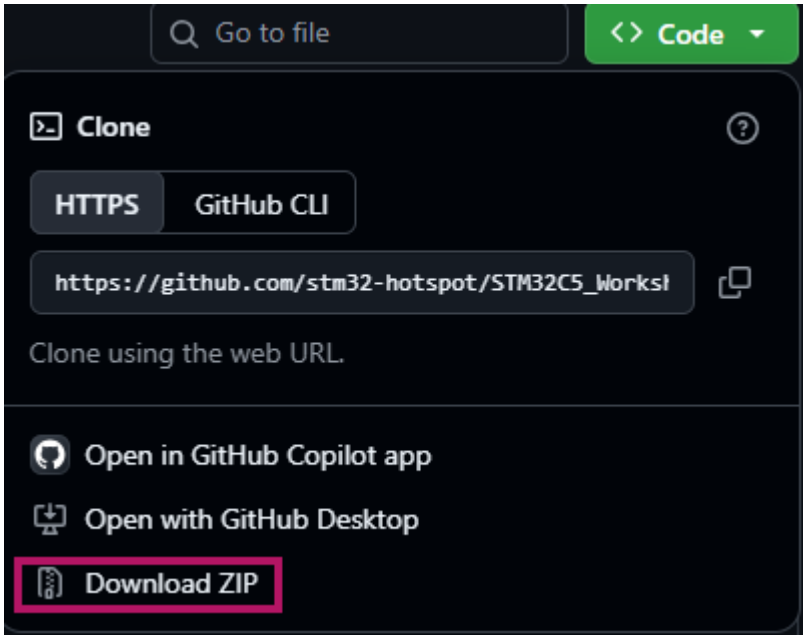
>cube mx pack-manager install --pack-ids STMicroelectronics.CMSIS.6.3.0 STMicroelectronics.stm32c5xx_dfp.2.0.0 STMicroelectronics.stm32c5xx_hal_drivers.2.1.0 STMicroelectronics.stm32cubemx2_templates_no_os_stm32c5.2.1.0 STMicroelectronics.syscalls.2.0.0

Upon successful completion of the install you should see this message:

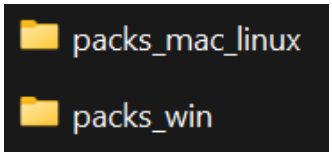
```
[Pack Manager CLI] - Installation completed in 0.114s
{
  "status": "success",
  "data": [
    "STMicroelectronics.CMSIS.6.3.0",
    "STMicroelectronics.stm32c5xx_dfp.2.0.0",
    "STMicroelectronics.stm32c5xx_hal_drivers.2.1.0",
    "STMicroelectronics.stm32cubemx2_templates_no_os_stm32c5.2.1.0",
    "STMicroelectronics.syscalls.2.0.0"
  ]
}
```

3. From Github – direct download

You can download the .zip file from this Github Repository: [stm32-hotspot/STM32C5_Workshop]



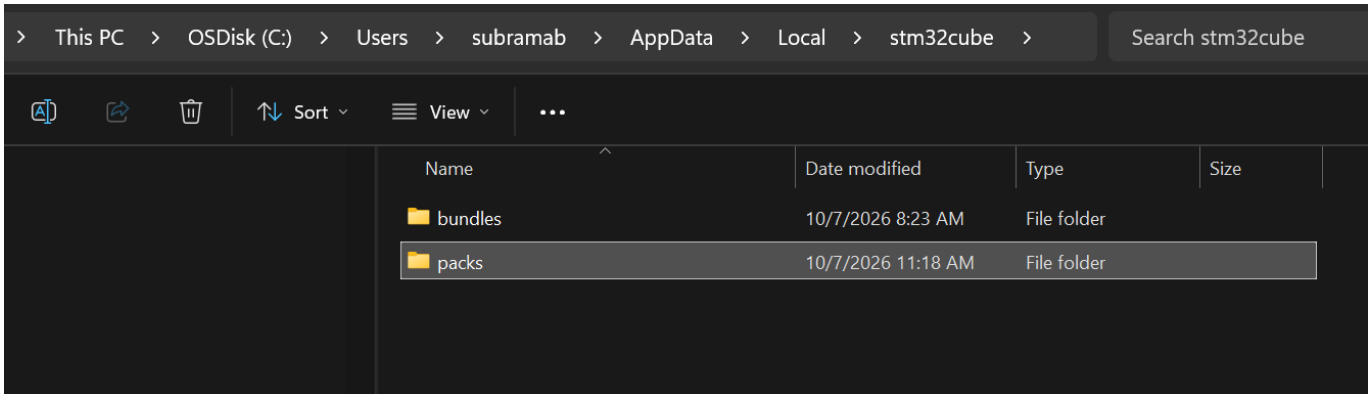
To install the packs, first find the packs that correspond to your operating system.



Move the contained **packs** folder into the appropriate location for your operating system

- **Windows**: 'C:\Users\<mylogin>\AppData\Local\stm32cube\packs'
- **Linux:** 'Home/.local/share/stm32cube/packs'
- **macOS:** 'Users/<mylogin>/Library/Application Support/stm32cube/packs'

Example on Windows Operating System



2- VSCode with STM32CubeIDE Extension

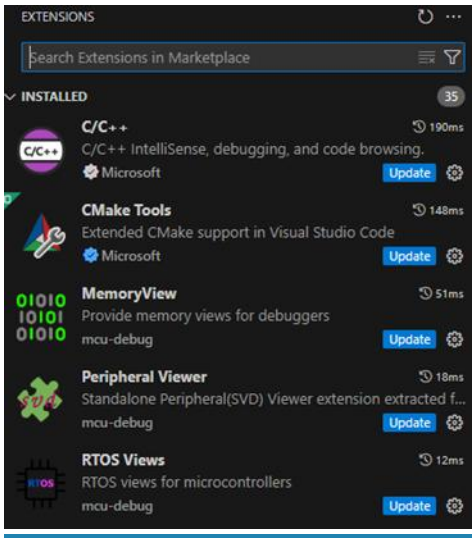
Click Link: [Installing Visual Studio Code on Windows]
Or use Direct Install Link:
https://code.visualstudio.com/docs/setup/windows#_install-vs-code-on-windows

Follow the installation provide in the link above:

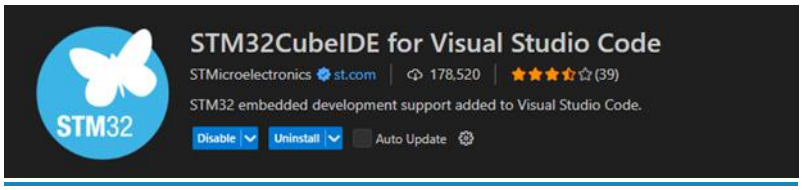
Setting up the STM32 VS Code Extension

First, you'll need to download the STM32VSCode extension. This is done through the VS Code application with your PC connected to the internet.

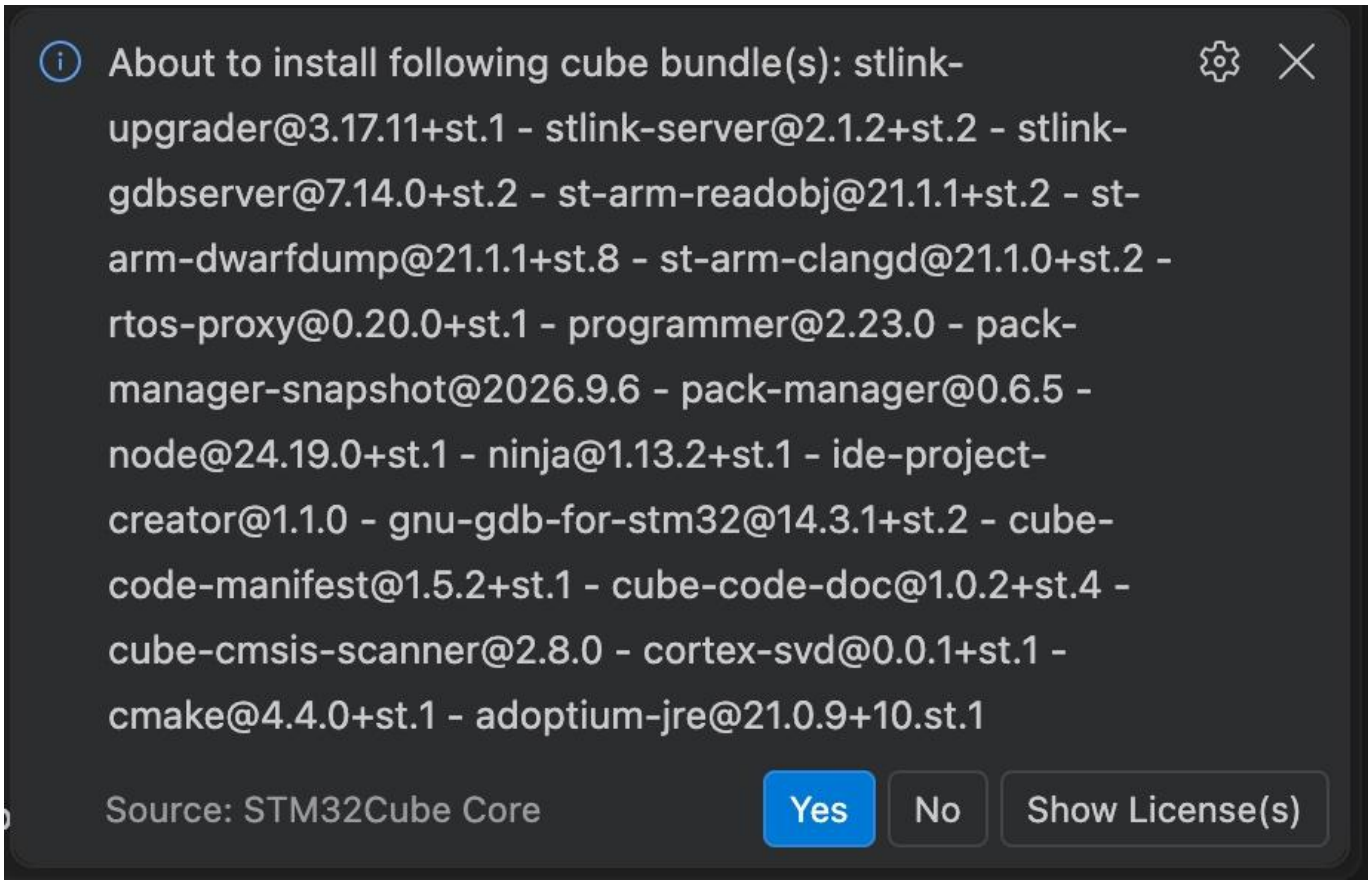
1. In the VS Code application, open **[Extensions]** (Ctrl + Shift + X).



2. Install the **[STM32 VS Code Extension].** It will look like this picture below.



3. After the installation, you may see a dialog box appear on the bottom right of the screen asking to install bundles. Here you can select **Yes**

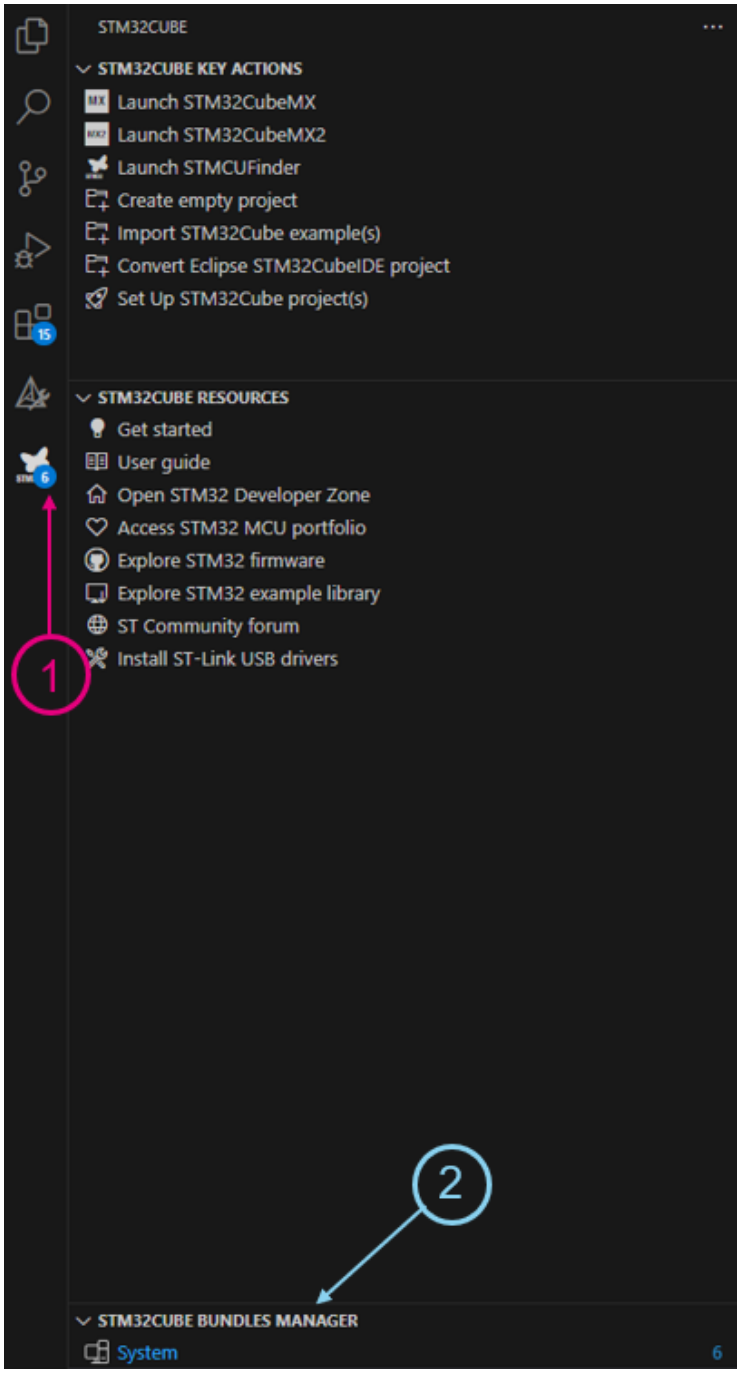


4. You will also have to install the following bundles manually
    - gnu-tools-for-stm32
    - gnu-tools-for-stm32-14_3_1-description

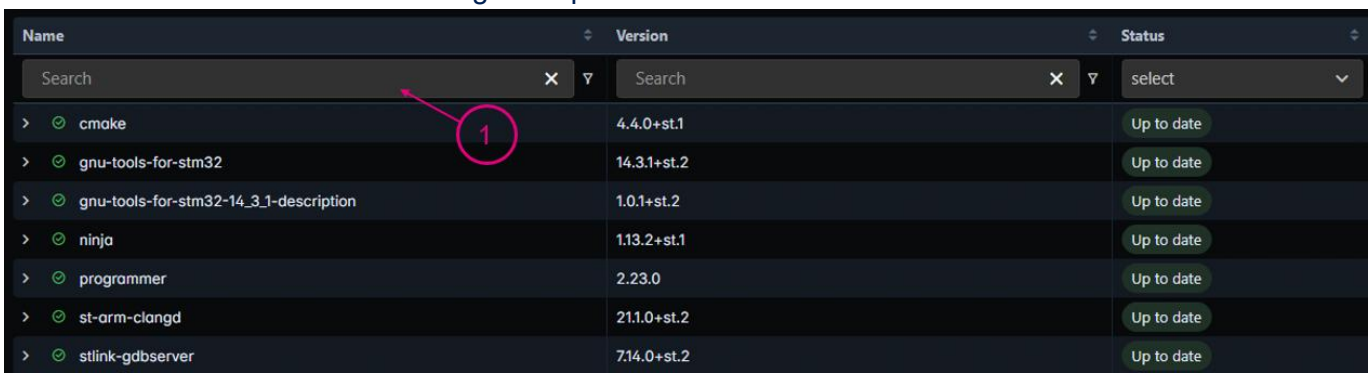
Installing the bundles will have to be done before you can configure a STM32 Project for the VSCode Extension. Installing bundles can be done in three different ways:

1. From Bundle Manager

You can access the Bundle Manager by clicking on the STM32CubeIDE for VSCode Extension, and then clicking Bundle Manager in the bottom right.


STM32CUBE
STM32CUBE KEY ACTIONS
Launch STM32CubeMX
Launch STM32CubeMX2
Launch STMCUFinder
Create empty project
Import STM32Cube example(s)
Convert Eclipse STM32CubeIDE project
Set Up STM32Cube project(s)
STM32CUBE RESOURCES
Get started
User guide
Open STM32 Developer Zone
Access STM32 MCU portfolio
Explore STM32 firmware
Explore STM32 example library
ST Community forum
Install ST-Link USB drivers
1
2
STM32CUBE BUNDLES MANAGER
System
6

You can search for the bundles using the top toolbar



Click the button on the far right of the bundle name to install the bundle.



5. You can download all the necessary bundles manually by searching in the Bundle Manager.

   The required bundles are:
   - adoptium-jre
   - cmake
   - cortex-svd
   - cube-cmsis-scanner
   - cube-code-doc
   - cube-code-manifest
   - gnu-gdb-for-stm32
   - gnu-tools-for-stm32
   - gnu-tools-for-stm32-14_3_1-description
   - ide-project-creator
   - ninja
   - pack-manager-snapshot
   - programmer
   - rtos-proxy
   - st-arm-clangd
   - st-arm-dwarfdump
   - st-arm-readobj
   - stlink-gdbserver

- stlink-server
- stlink-upgrader

2. *From Command Line*

You can use the following command to download the necessary bundles from the CLI.

```
>cube bundle install adoptium-jre@21.0.9+10.st.1 cmake@4.4.0+st.1
cortex.svd@0.0.1+st.1 cube-cmsis-scanner@2.8.0 cube-code-doc@1.0.2+st.4 cube-
code-doc@1.0.2+st.4 cube-code-manifest@1.5.2+st.1 gnu-gdb-for-
stm32@14.3.1+st.2 gnu-tools-for-stm32@14.3.1_st.1 gnu-tools-for-stm32-14_3_1-
description@1.0.1+st.2 ide-project-creator@1.1.0 ninja@1.13.2+st.1 pack-
manager-snapshot@2026.96 programmer@2.23.0 rtos-proxy@0.20.0+st.1 st-arm-
clangd@21.1.0 st-arm-dwarfdump@21.1.1+st.8 st-arm-readobj@21.1.1+st.2 stlink-
gdbserver@7.14.0+st.2 stlink-server@2.1.1+st.8 stlink-upgrader@3.17.11+st.1
```

When you see the following message, reply ‘yes’

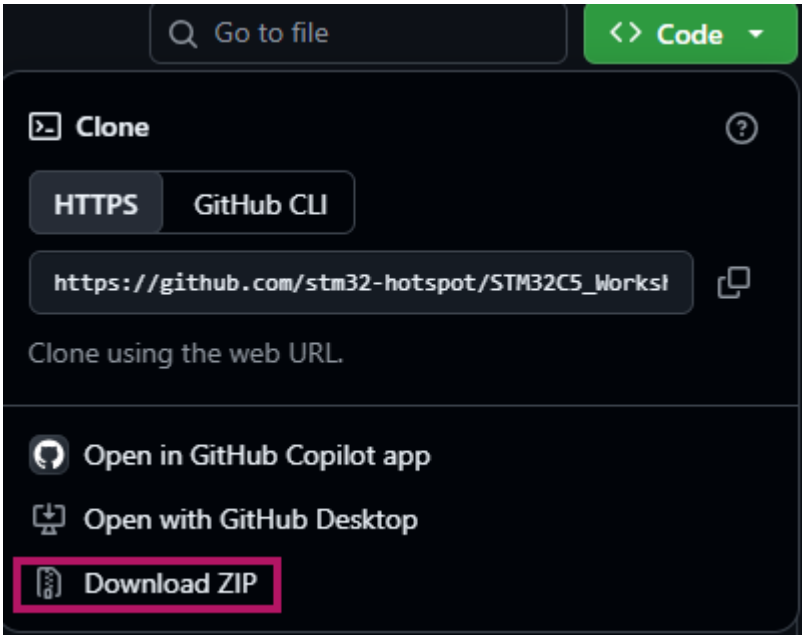
```
Do you want to install the following bundles or show the license(s): ["st-arm-clangd@19.1.2+st.3"] (yes, no, show)?
```

Once the bundles are installed you should see a message like the one below:

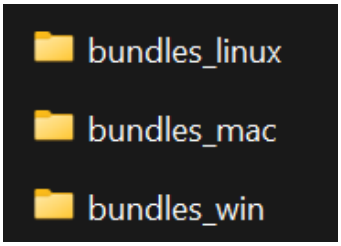
```
Installed 2 bundle(s)
Installation has been completed.
```

3. *From Github Repo*

You can download the .zip file from this Github Repository: [stm32-hotspot/STM32C5_Workshop]



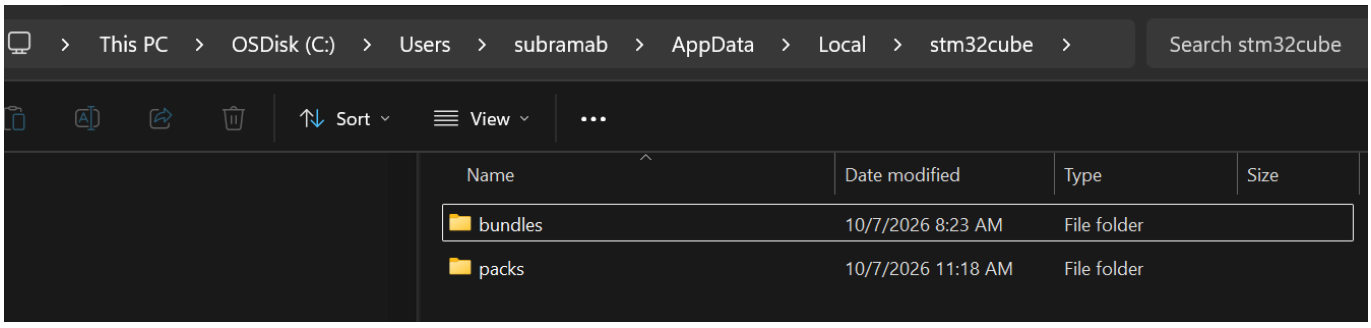
To install the bundles, first find the bundles that correspond to your operating system.



Move the contained **bundles** folder into the appropriate location for your operating system

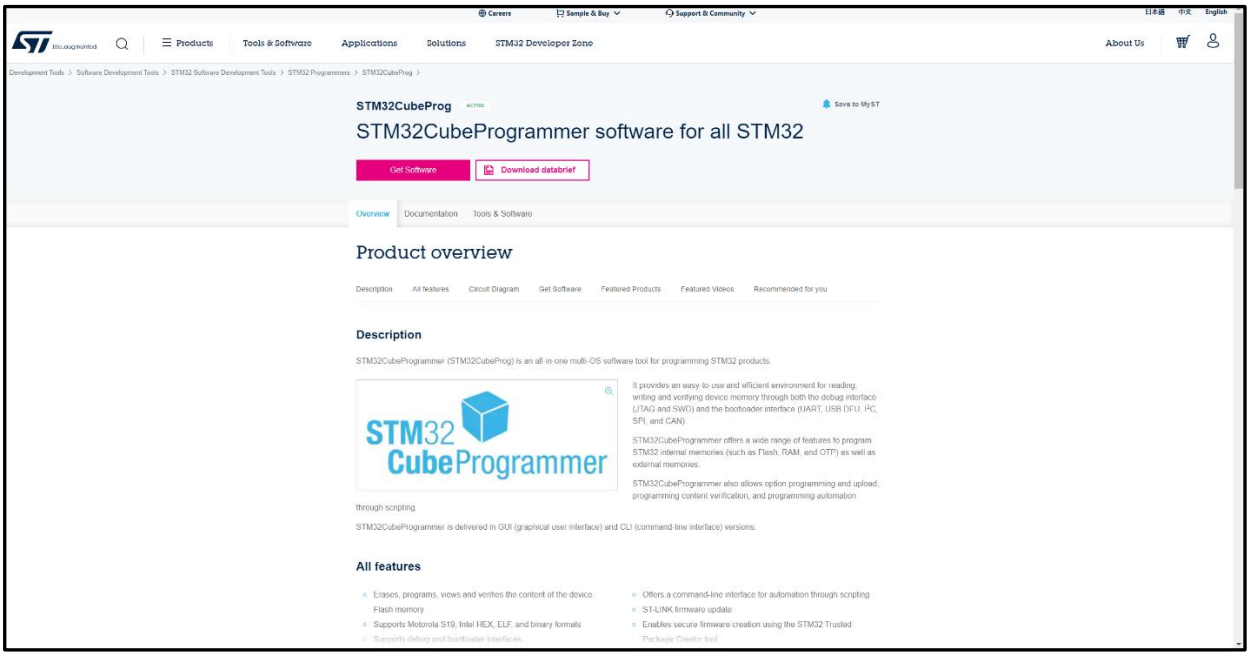
- **Windows**: 'C:\Users\<mylogin>\AppData\Local\stm32cube\bundles
- **Linux:** 'Home/.local/share/stm32cube/bundles
- **macOS:** 'Users/<mylogin>/Library/Application Support/stm32cube/bundles

Example on Windows Operating System



3 - STM32CubeProg: STM32CubeProgrammer software for programming STM32 products

Click Link: STM32CubeProg
Minimum version required: 2.23.0
Direct Install Link:
https://www.st.com/content/st_com/en/products/development-tools/software-development-tools/stm32-software-development-tools/stm32-programmers/stm32cubeprog.html



Click on: 



Please get the latest software version available that will be accessible by clicking on:

Get latest

License Agreement

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BY CLICKING ON THE "I ACCEPT" BUTTON OR BY UNZIPPING, INSTALLING, COPYING, DOWNLOADING, ACCESSING OR OTHERWISE USING THIS SOFTWARE PACKAGE OR ANY PART THEREOF, INCLUDING ANY RELATED DOCUMENTATION (collectively the “SOFTWARE PACKAGE”) FROM STMICROELECTRONICS INTERNATIONAL N.V, SWISS BRANCH AND/OR ITS AFFILIATED COMPANIES (collectively "STMICROELECTRONICS”), YOU (hereinafter referred also to as “THE RECIPIENT”), ON BEHALF OF YOURSELF, OR ON BEHALF OF ANY ENTITY BY WHICH YOU ARE EMPLOYED AND/OR ENGAGED, AGREE TO BE BOUND BY THIS AGREEMENT.

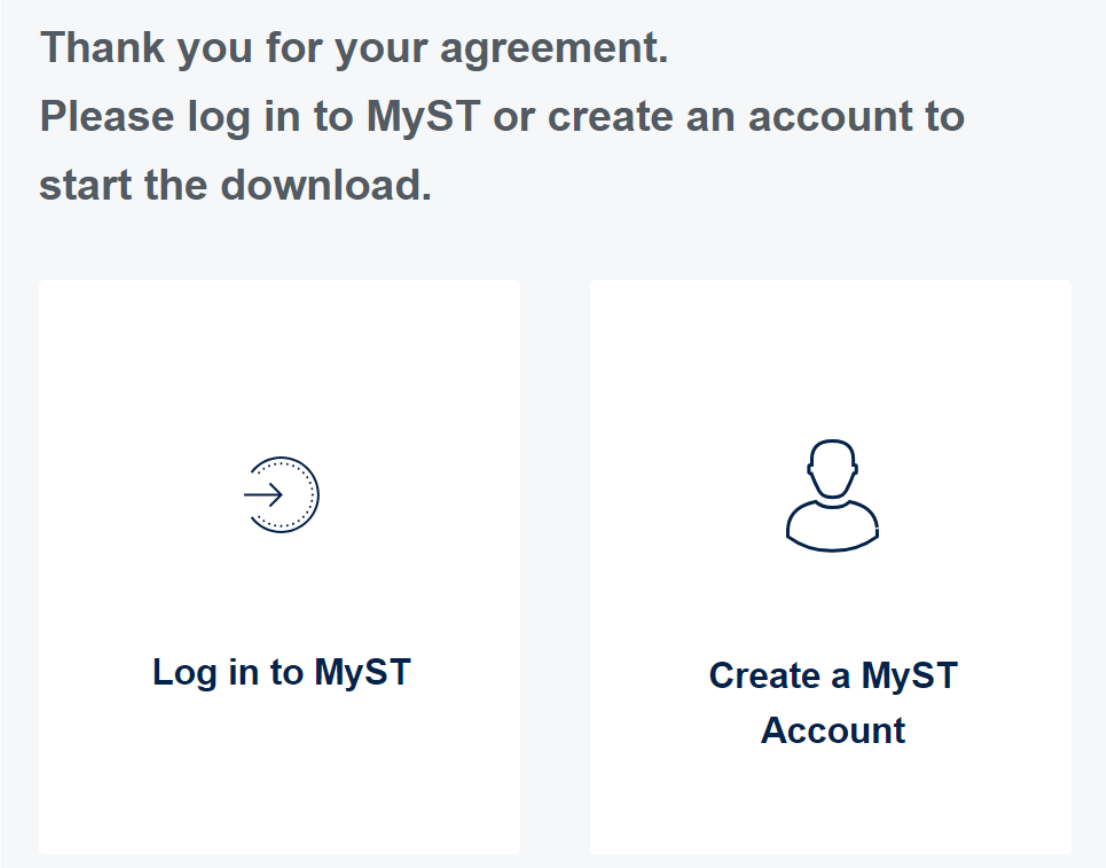
Under STMICROELECTRONICS’ intellectual property rights and subject to applicable licensing terms for any third-party software incorporated in the SOFTWARE PACKAGE and applicable Open Source Terms (as defined here below), the redistribution, reproduction and use in source and binary forms of the SOFTWARE PACKAGE or any part thereof, with or without modification, are permitted provided that the following conditions are met:

1. Redistribution of source code (modified or not) must retain any copyright notice accompanying the SOFTWARE PACKAGE, this list of conditions and the disclaimer below.

Additional License Terms for STM32CubeProgrammer 2.23.0

Decline Accept

Click on: Accept



There are 3 ways of getting the software:

If you have an ST account, click on: **Log in to MyST**

If you don’t have one, create one by clicking on: **Create a MyST Account**

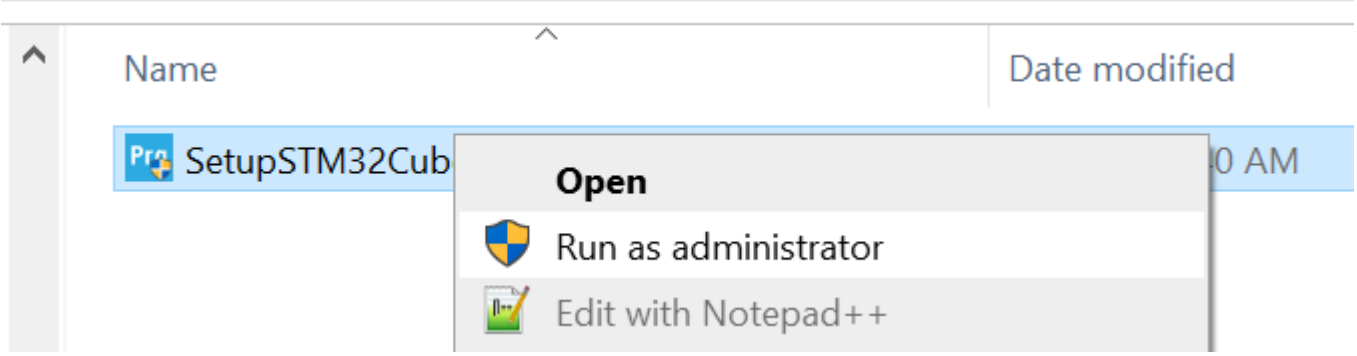
If you don’t have an account and don’t want to create one, Click “”Download as a guest”

Then click: Download latest

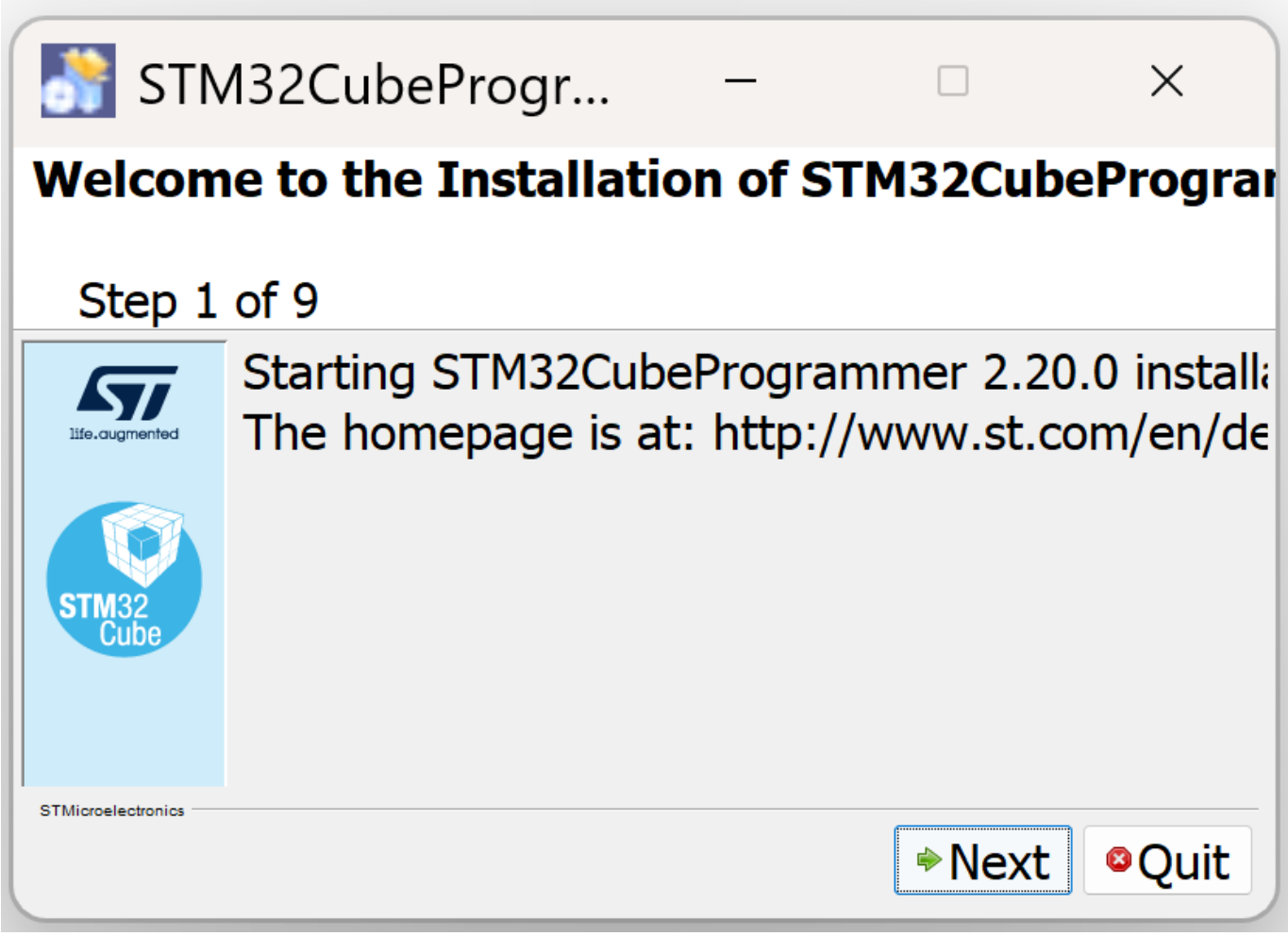
Unzip the file (SetupSTM32CubeProgrammer_win64)

Note: you may see a newer version in your case than the screenshot above.

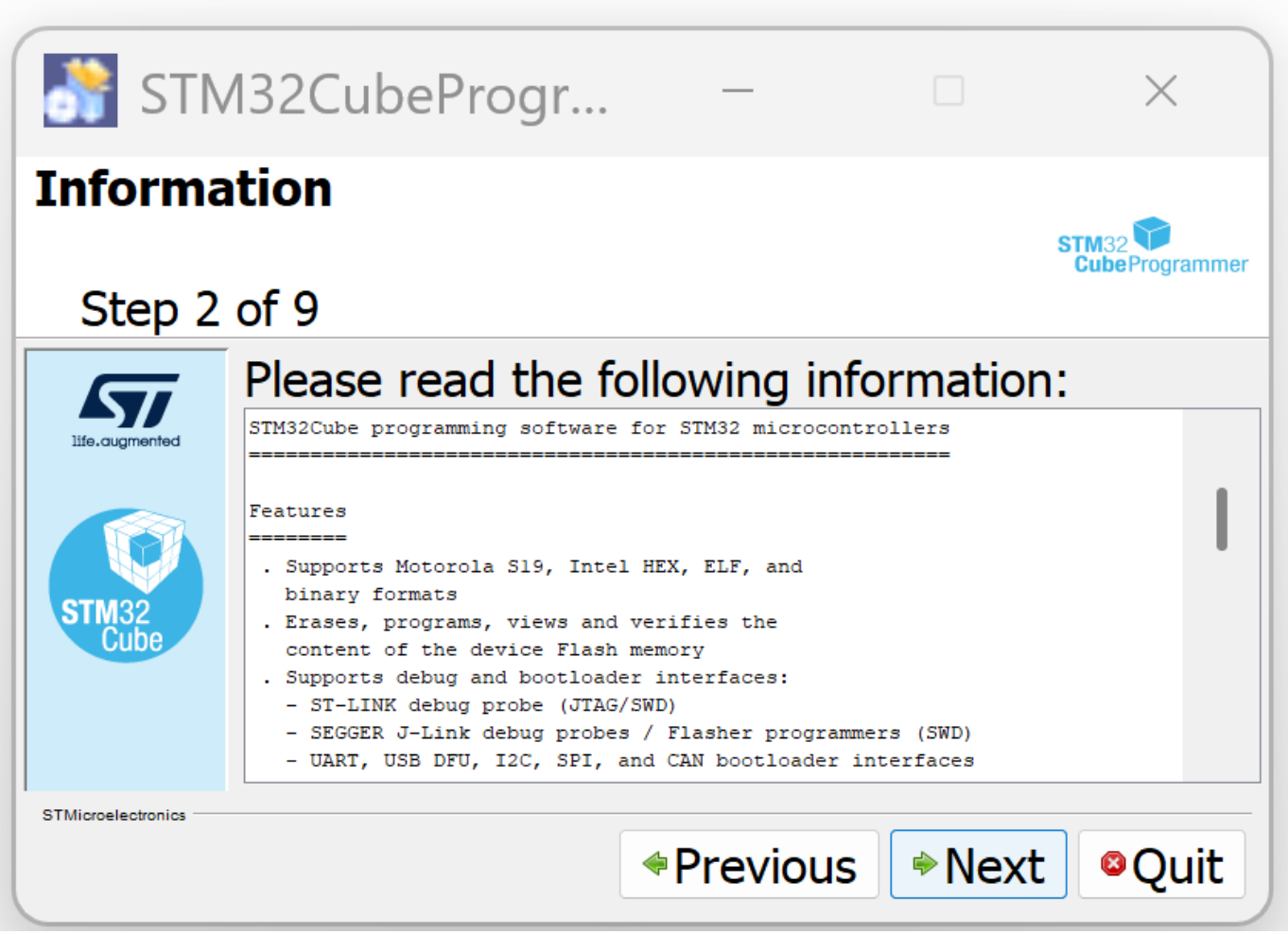
Right Click on SetupSTM32CubeProgrammer-win64.exe and Run as administrator.



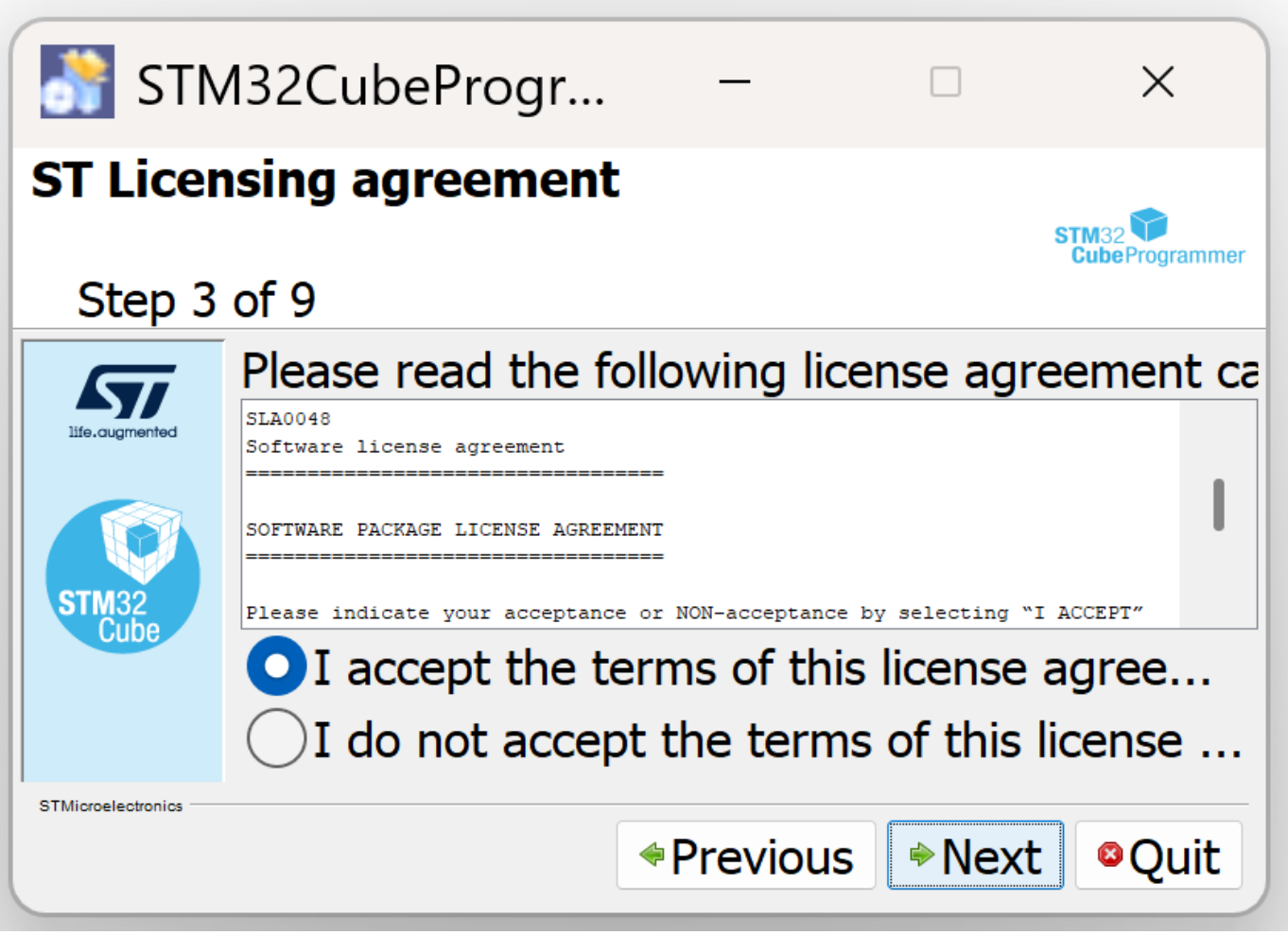
Press Next:



Press Next:



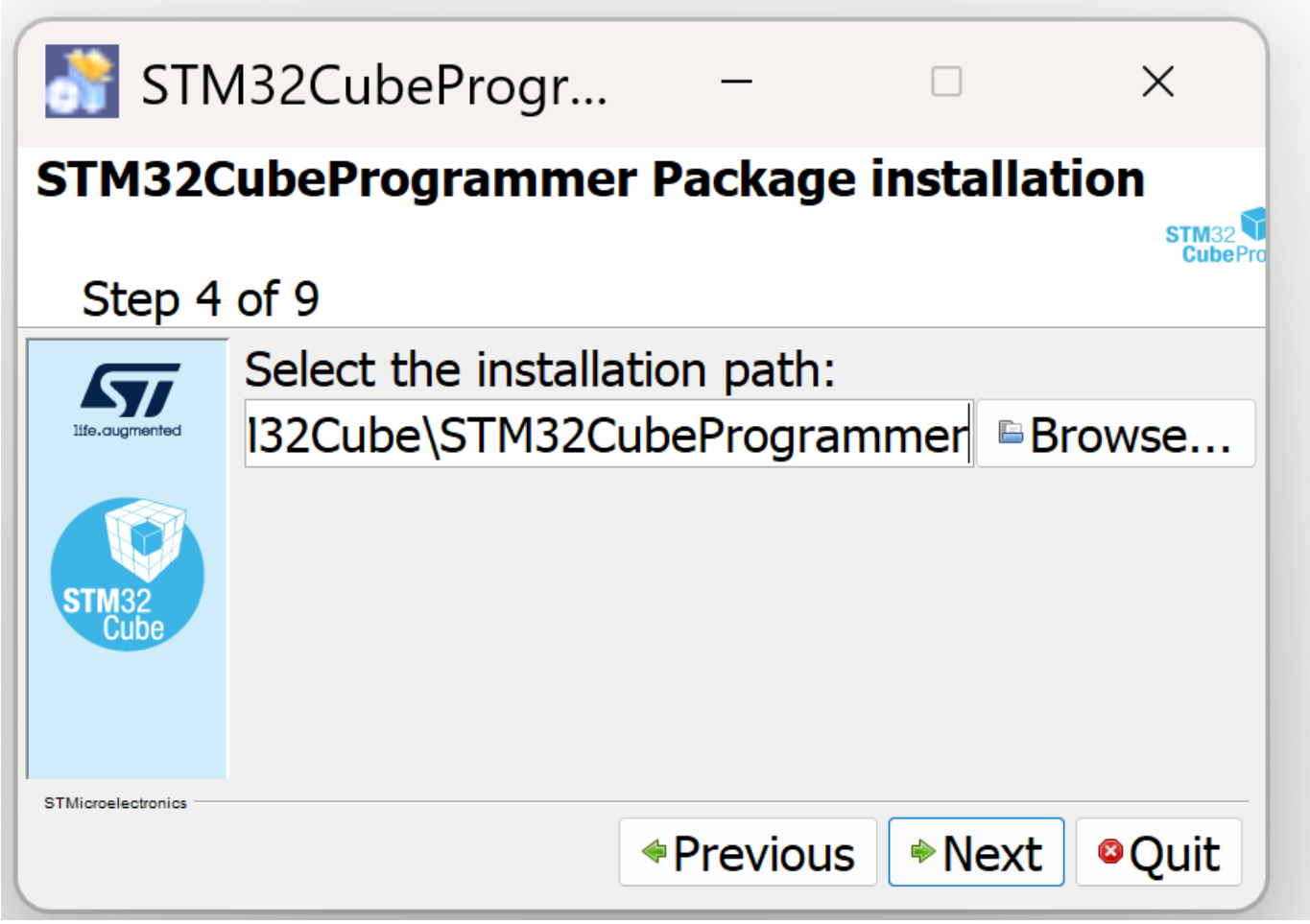
Accept the Terms and then click on Next:



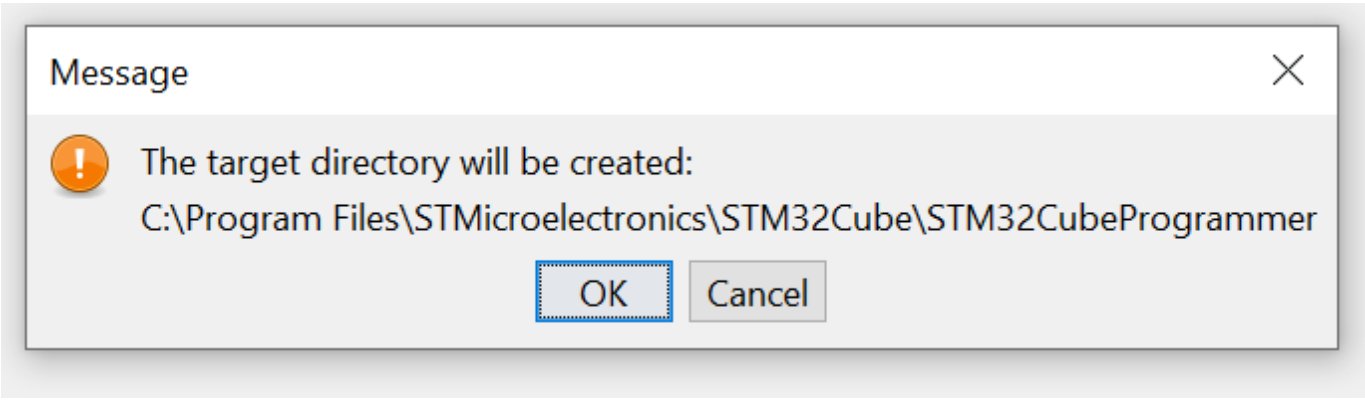
Accept the Privacy Policy and Terms of Use

Important note: If you have a previous version, it will be overwritten if you keep the default path. So, if you want to keep your previous version you will need to install the new version in a different directory. For the workshop, be sure to use the version that we are installing here or a later one.

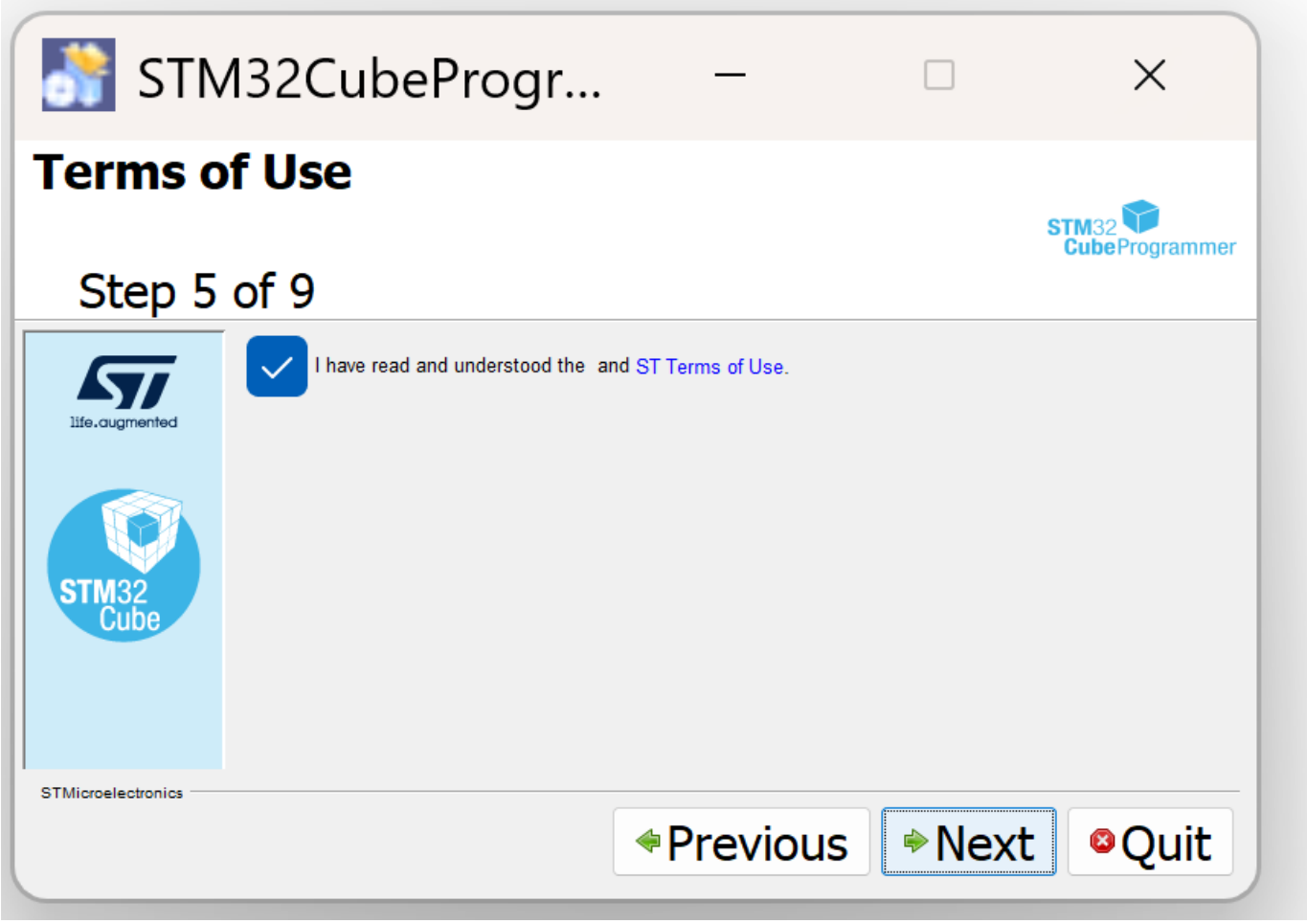
Once you have decided on the path then press “Next”:



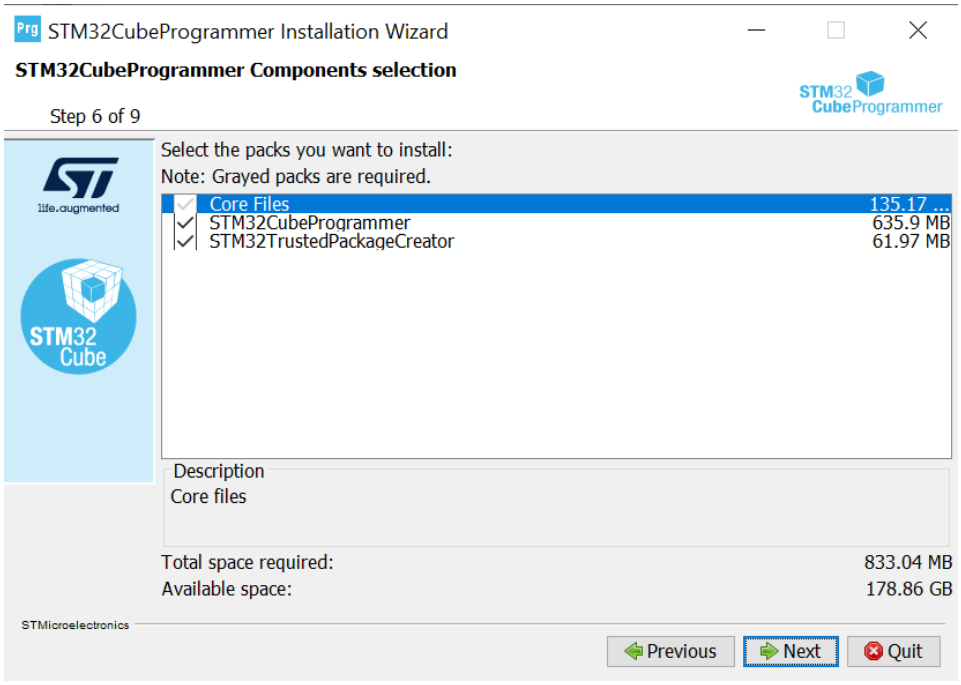
Click OK:



Acknowledge and click Next:

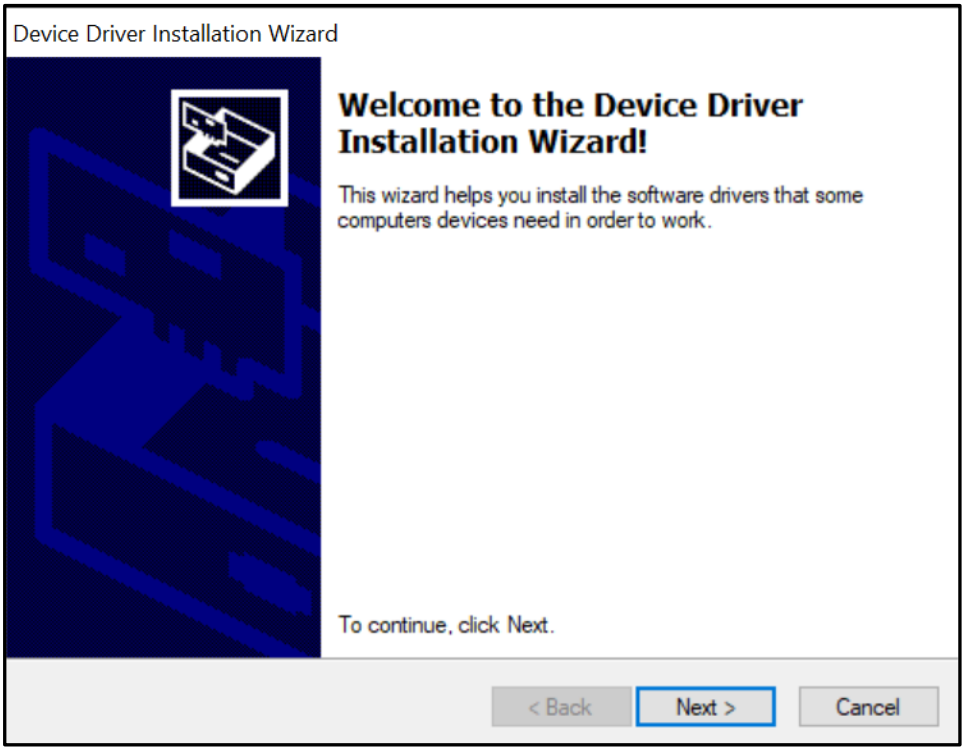


Select the packs that you want to install and press “Next”:



Follow those steps to install the ST-LINK Driver:

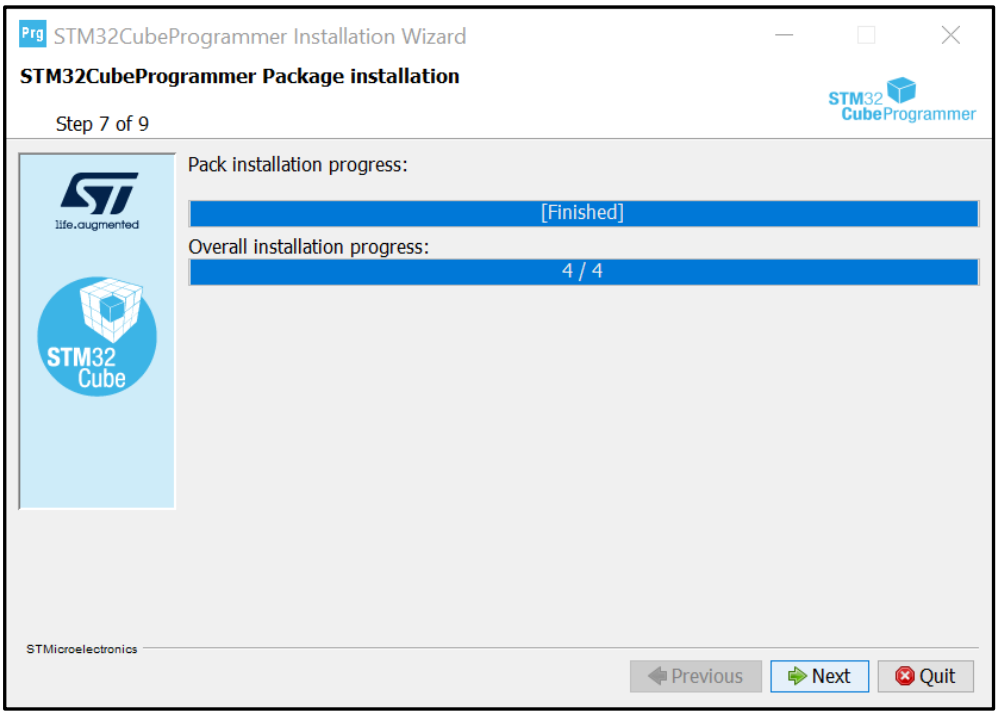
Press Next:



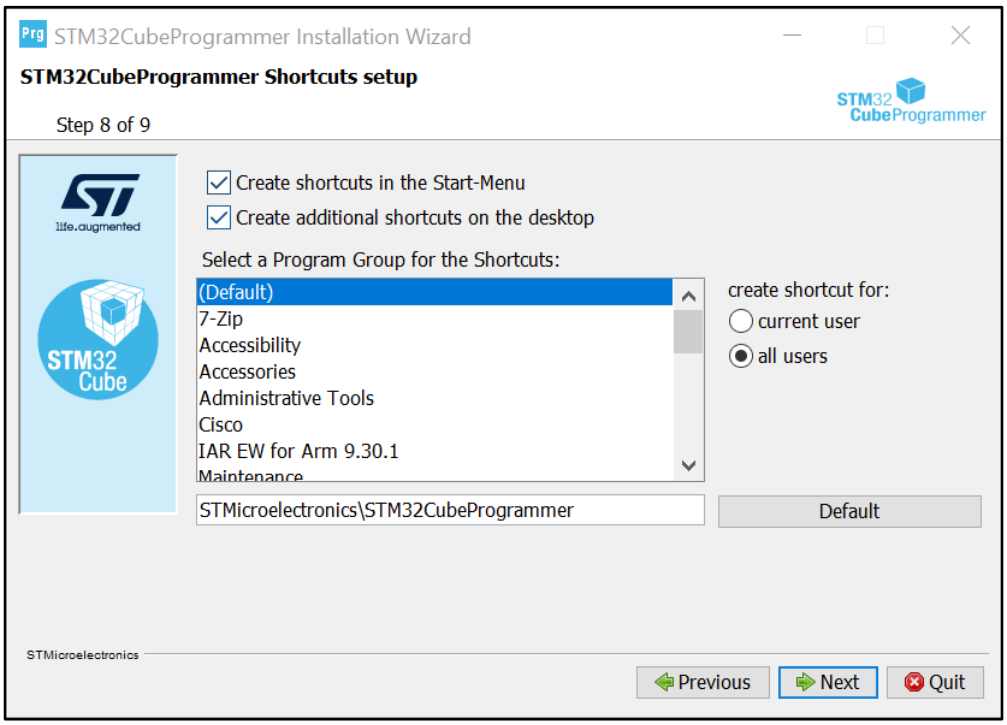
Press Finish:



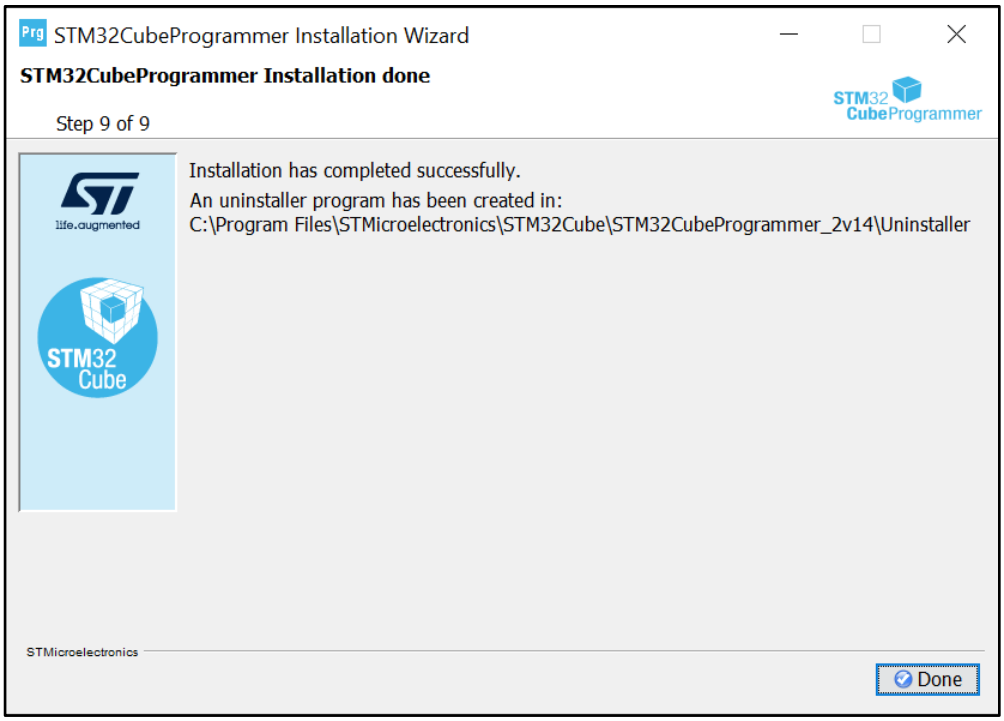
Wait for the install to complete and then press Next:



Select the options to create the shortcuts as desired and then press next:



Press Done:



Open STM32CubeProgrammer to make sure it was installed properly.

4 – Presentation and code to be added during the workshop:

The code from lab (hands-on exercises) will be included in the Code_to_add zip file. During the workshop, step-by-step instructions are included with regards to editing the starter code using the provided code segments. Rather than retyping the code, you will only need to do a simple copy-paste operation.